

Study Notes: Introduction to Python

Programming ### 1. Overview of Python *

****Definition:**** Python is a high-level, interpreted programming language. *

****Key Characteristics:**** Known for being simple, easy to learn, powerful, and flexible. *

****History:**** Created by ****Guido van Rossum**** and first released in

****1991****. --- ### 2. Key Advantages *

****Simple Syntax:**** Python's clean and readable code structure makes it an ideal language for beginners. *

****Efficiency:**** Features a large collection of libraries that allow developers to build applications quickly.

* ****Versatility:**** Supports multiple programming paradigms, including: *

****Object-Oriented Programming (OOP)****

* ****Functional Programming**** *

****Procedural Programming**** --- ### 3.

Real-World Applications Python is widely used across various high-growth

industries: * ****Web Development**** *

****Artificial Intelligence (AI)**** *

****Data Science**** * ****Machine Learning****

* ****Automation**** * ****General Software**

Development** --- ### 4. Web Development

Frameworks Frameworks are tools that simplify the process of building web

applications. Popular examples include: *

****Flask:**** A lightweight framework often

used for developing AI-integrated websites. * **Django:** (Commonly referred to as "Mongo" in the lecture notes; standard industry practice typically uses Django alongside Flask).

--- ## Exam Quick-Review Summary /

Concept / Key Point / / :--- / :--- / /

Language Type / High-level,

interpreted. / / **Creator** / Guido van

Rossum (1991). / / **Why it's popular** /

Simple syntax, large library support, and

flexibility. / / **Primary Use Cases** /

AI, Data Science, Web Dev, Automation. / /

Key Framework / Flask (used for AI

nodes and website generation). / ***

Study Tip: *When preparing for exams,

focus on why Python is considered

"beginner-friendly" (simple syntax) and be

prepared to list at least three areas

where Python is used (e.g., AI, Data

Science, Web Dev). *